



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr.Jayaram Boga
Student Name	M.V.L.Samhita
Roll Number	2520030509
Branch	CSE
Course Outcome	CO2

Case Study Topic: Smart Irrigation Water Level Monitoring using Segment Tree with Lazy Propagation

Work Done Summary:

In this case study, we implemented a Segment Tree with Lazy Propagation for monitoring water levels across agricultural irrigation zones.

The Segment Tree stores water level values in different irrigation zones and supports efficient range update and range maximum query operations.

We inserted multiple water level updates into the Segment Tree and observed lazy propagation for optimized interval processing.

We also implemented range maximum querying in Java to efficiently identify the highest water level within selected irrigation regions.

Implementation Details:

1. Initialized 16 irrigation zones with water level value 1.0.
2. Constructed Segment Tree for efficient range maximum operations.
3. Implemented Lazy Propagation for optimized range addition updates.
4. Applied multiple water level update operations on irrigation intervals.
5. Performed maximum water level queries on selected ranges.
6. Concluded Segment Tree with Lazy Propagation is efficient for large-scale interval update and query operations.

Result: Screenshots / Output

```
Initial Water Levels:
1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0

Maximum Water Level in Zones [0,15] : 1.8

Maximum Water Level in Zones [4,10] : 2.2
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Time Complexity:

Segment Tree Range Update (Lazy Propagation): $O(\log n)$

Segment Tree Range Maximum Query: $O(\log n)$

Compared to arrays and brute-force traversal, Segment Trees with Lazy Propagation provide significantly better performance for large-scale irrigation monitoring systems.

GitHub Link: <https://github.com/venkatalaxmisamhita/DSA-II-PROJ>

Student Signature	Faculty Signature